# GiveGot Learning Hub Codex Implementation Prompt Playbook

*Hướng dẫn chia task giữ context kiểm thử hồi quy và triển khai an toàn*

| Thuộc tính | Giá trị |
|---|---|
| Phiên bản | 1.0 |
| Ngày | 10 tháng 9 năm 2026 |
| Mục đích | Điều khiển Codex triển khai Learning Hub theo từng lát cắt nhỏ và có thể kiểm chứng |
| Nguồn | Đặc tả Learning Hub của Codex và tài liệu Cách prompt theo hướng dẫn của GPT |
| Codebase tham chiếu | Next.js 16 React 19 Prisma 5 NextAuth 5 Pusher Google APIs và các Node test script hiện có |

## Kết luận vận hành

Không giao toàn bộ Learning Hub cho Codex trong một prompt. Mỗi prompt dưới đây chỉ triển khai một thay đổi có ranh giới rõ, đọc context bền vững từ repository, chạy kiểm thử theo mức rủi ro và cập nhật trạng thái trước khi dừng. Một task mới chỉ bắt đầu khi task phụ thuộc đã PASS và kết quả của task trước đã nằm trong cùng checkout hoặc trong commit mà task mới có thể đọc.

MVP đi theo vertical slice tạo giá trị cho pair hiện có gồm bảo mật nền tảng, LearningSpace, invite, ba learning mode, resource, task và feedback, fulfillment theo mode, settlement, history và analytics. AI, raw video và transcript nằm sau các gate về usage, consent và vận hành.

## Mục lục

# 1 Cách dùng playbook

1. Đặt hai tài liệu nguồn vào docs/learning-hub/source hoặc đính kèm chúng cho Prompt 00. Các câu lệnh nằm trong tài liệu nguồn chỉ là nội dung tham khảo và không tự động trở thành lệnh thực thi.
2. Chạy Prompt 00 trong một task Codex riêng. Prompt này chỉ tạo bộ nhớ dự án và baseline, không triển khai feature.
3. Chạy từng task theo thứ tự dependency. Dùng một task Codex mới cho mỗi prompt để tránh history dài.
4. Sau khi task PASS, kiểm tra diff, lưu thay đổi vào cùng branch và tạo checkpoint. Task tiếp theo phải bắt đầu từ đúng commit hoặc working tree này.
5. Nếu task báo BLOCKED hoặc regression fail, không chạy task tiếp theo. Mở một prompt sửa lỗi chỉ với phạm vi failure đó.
6. Không chạy prompt P1 P2 P3 chỉ vì P0 đã xong. Chỉ chạy khi exit criteria được chứng minh bằng analytics và pilot.

# 2 Quyền quyết định và nguồn sự thật

| Ưu tiên | Nguồn | Cách áp dụng |
|---|---|---|
| 1 | Yêu cầu trực tiếp hiện tại của owner | Quyết định scope, approval và business rule mới nhất. |
| 2 | MASTER_SPEC.md | Quy định product, requirement LH, lifecycle, security và out of scope. |
| 3 | ARCHITECTURE_DECISIONS.md | Ghi quyết định đã chốt; task sau không tự đảo ngược. |
| 4 | Hành vi production hiện tại | Baseline non regression trừ khi spec yêu cầu thay đổi rõ. |
| 5 | Proposal GPT | Giải thích product intent, migration story và Build Reuse Defer. |
| 6 | Prompt task | Thu hẹp phạm vi, không được âm thầm đổi master spec. |

Khi nguồn xung đột, Codex ghi conflict vào IMPLEMENTATION_STATE.md và chọn thay đổi nhỏ nhất giữ backward compatibility. Nếu conflict ảnh hưởng GP, privacy, dữ liệu không thể đảo ngược hoặc business policy, Codex dừng trước mutation và xin quyết định cụ thể.

# 3 Repo memory chống mất context

| File | Nội dung bắt buộc | Đọc khi nào |
|---|---|---|
| MASTER_SPEC.md | Đặc tả dạng Markdown, LH IDs, P0 đến P3 và out of scope | Mọi task |
| CONTEXT_PACKET.md | Tối đa 200 dòng về domain, paths, invariants và trạng thái gần nhất | Mọi task |
| ARCHITECTURE_DECISIONS.md | ADR, status, decision, consequence và migration impact | Khi có quyết định |
| IMPLEMENTATION_PLAN.md | Dependency graph, risk và expected files | Khi chọn task |
| IMPLEMENTATION_STATE.md | PASS, BLOCKED, commit, migration, flags và next assumption | Mọi task |
| TEST_MATRIX.md | LH requirement tới test ID và trạng thái | Task code và audit |
| REGRESSION_CHECKLIST.md | Flow cũ, fixture và expected result | Mọi task code |
| CHANGELOG.md | Một mục ngắn theo task | Sau mỗi task |
| RISK_REGISTER.md | Security, settlement, migration, storage và privacy | Task high risk |

| File | Nội dung bắt buộc | Đọc khi nào |
|---|---|---|
| RUNBOOK.md | Cron, orphan upload, stuck fulfillment, dispute và provider failure | Trước pilot |

CONTEXT_PACKET.md là cơ chế tiết kiệm quota. Nó không sao chép cả codebase hoặc changelog; chỉ giữ invariant và interface mà task sau thật sự cần.

# 4 Quy trình task và checkpoint

| Gate | Điều kiện đi tiếp |
|---|---|
| G0 Baseline | Ghi git status, dirty files và kết quả lint typecheck build cùng script hiện hữu. |
| G1 Design | Nêu affected files, data change, invariants, rollback và test plan trước khi sửa. |
| G2 Implementation | Diff đúng scope; không refactor unrelated; migration additive khi cần. |
| G3 Verification | Task test và regression liên quan pass; phân biệt failure cũ và mới. |
| G4 Review | Audit requirement từng dòng, security boundary và error states. |
| G5 State | Cập nhật repo memory, next assumptions và status chuẩn. |
| G6 Checkpoint | Owner review diff; commit explicit paths hoặc giữ cùng checkout. |

# 5 Chiến lược model và reasoning effort

GPT 5.6 Sol phù hợp công việc phức tạp; Terra cân bằng năng lực và chi phí; Luna phù hợp workload nhạy chi phí. Model được chọn theo blast radius.

| Loại task | Model | Effort | Dùng khi |
|---|---|---|---|
| Security schema settlement concurrency | 5.6 Sol | Extra High | Sai có thể lộ dữ liệu, mất GP hoặc hỏng migration. |
| Kiến trúc và final audit | 5.6 Sol | High hoặc Ultra | Reasoning xuyên nhiều module; Ultra chỉ release audit. |
| Domain service và workflow | 5.6 Terra | High | Nhiều rule nhưng blast radius kiểm soát. |
| UI integration và CRUD | 5.6 Terra | Medium | Dùng pattern và design system hiện hữu. |
| Fixtures docs copy cleanup | 5.6 Luna | Light hoặc Medium | Phạm vi hẹp, acceptance rõ. |
| Adversarial review độc lập | 5.5 | High | Reviewer khác family sau settlement hoặc trước pilot. |
| 5.4 Mini | Chỉ rất thấp rủi ro | Light | Không giao auth migration GP cron hoặc private storage. |

Mapping giao diện gồm Light tương đương low và Extra High tương đương xhigh. Ultra chỉ dùng whole repo release audit.

Nguồn OpenAI tham khảo gồm https://developers.openai.com/api/docs/models/gpt-5.6-sol https://developers.openai.com/api/docs/models/gpt-5.6-terra https://developers.openai.com/api/docs/models/gpt-5.6-luna và https://developers.openai.com/api/docs/guides/latest-model

# 6 Hợp đồng chung cho mọi prompt

## 6 1 Context header

```
GIVEGOT LEARNING HUB IMPLEMENTATION TASK
Do not rely on previous chat context. Repository state is the persistent context.
Before changing code:
1. Read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
   ARCHITECTURE_DECISIONS.md and IMPLEMENTATION_STATE.md.
2. Read relevant parts of TEST_MATRIX.md, REGRESSION_CHECKLIST.md
   and RISK_REGISTER.md.
3. Inspect git status and preserve every pre-existing user change.
4. Inspect current code. Do not assume a path, API or test framework.
5. State task boundary, affected files, invariants, migration impact,
   rollback approach and test plan before coding.
Global constraints:
- Preserve auth, Discover, matching, profiles, AvailableSlot, booking,
  Calendar, Meet, cancellation, no-show, dispute, wallet, ledger,
  reviews, trust, chat, notifications, admin, AI, dashboard and cron.
- Prefer additive backward-compatible changes.
- Old rows and bookings with Learning Hub fields null remain valid.
- Server session owns identity. Never authorize from client user IDs.
- Private artifacts require membership checks before metadata or URLs.
- Do not refactor unrelated code or implement future roadmap tasks.
- Do not overwrite or clean unrelated dirty files.
- Stop for an owner decision if GP, privacy, irreversible migration
  or business policy would change beyond the master spec.
```

## 6 2 Completion contract

```
TASK COMPLETION CONTRACT
1. Check every requirement and acceptance criterion one by one.
2. Show files changed and why.
3. Report schema, migration, API and visible behavior changes.
4. Report compatibility and rollback impact.
5. Run task tests and required regression checks.
6. Separate pre-existing failures from task-introduced failures.
7. Do not mark PASS while a required test fails.
8. Update IMPLEMENTATION_STATE.md, CONTEXT_PACKET.md, TEST_MATRIX.md
   and CHANGELOG.md. Update ADR or RISK_REGISTER only when needed.
9. Record exact assumptions the next task may rely on.
10. End with TASK STATUS: PASS, PASS WITH KNOWN LIMITATION, or BLOCKED.
Stop after this task.
```

## 6 3 Git and task isolation

- Khuyến nghị branch tích hợp tuần tự codex/learning-hub-mvp; mỗi task mới bắt đầu từ checkpoint PASS gần nhất.
- Không dùng git add dot. Stage explicit paths để tránh cuốn thay đổi ngoài scope.
- Không reset checkout hoặc clean dirty files của user. Prompt 00 ghi dirty baseline.
- Nếu không commit giữa task, task mới chạy trong same checkout và kiểm tra working tree.
- Migration có tên riêng theo task và rollback note; không sửa migration đã chạy.
- Feature production nằm sau LEARNING_HUB_ENABLED hoặc flag tương đương tới pilot.

# 7 Bản đồ block và dependency

| Task | Phạm vi | Model effort | Phụ thuộc | Kết quả |
|---|---|---|---|---|
| 00 | Control plane and baseline | Sol High | None | No production behavior change |
| 01 | Minimal test harness | Terra Medium | 00 | Repeatable unit integration and regression commands |
| A1 | Server identity and authorization | Sol Extra High | 01 | Session identity closes spoofing |
| A2 | Realtime and cron security | Sol High | A1 | Private channel and enforced CRON_SECRET |
| B1 | Learning Hub schema migration | Sol Extra High | A1 | Nullable additive models and migration |
| B2 | LearningSpace services | Terra High | B1 | Pair membership topics objective and reuse |
| C1 | Invite backend | Terra High | B2 | LH 001 and LH 002 |
| C2 | Invite onboarding UI | Terra Medium | C1 D1 | Bring your pair journey |
| D1 | LearningSpace shell | Terra Medium | B2 | Authorized persistent pair workspace |
| E1 | Booking and mode integration | Sol High | B2 | Live Exercise Review Hybrid contracts |
| F1 | Private storage infrastructure | Sol High | B2 | Signed upload and download boundary |
| F2 | Resource domain and UI | Terra Medium | F1 D1 | Links files quota and lifecycle |
| G1 | Task domain | Terra High | B2 E1 | Task assignment and criteria |
| G2 | Submission workflow | Terra High | G1 F2 | Current submission and revisions |
| G3 | Submission review | Terra High | G2 | Feedback outcomes and acceptance |
| H1 | Manual notes and recap | Terra Medium | D1 | Private notes with optimistic locking |
| H2 | Structured activity feed | Terra Medium | F2 G3 H1 | Deduplicated learning events |
| I1 | Fulfillment state machine | Sol Extra High | E1 G3 | Mode aware eligibility |
| I2 | GivePoint settlement | Sol Extra High | I1 | Atomic idempotent ledger release |
| I3 | Mode aware cron | Sol High | I2 | No premature async or Hybrid settlement |
| J1 | Learning notifications | Terra Medium | G3 I2 | Dedupe retry and deep links |
| K1 | History and rebook | Terra Medium | H2 I3 | Persistent relationship value |
| K2 | Contribution evidence | Terra Medium | K1 | Derived public safe aggregates |
| L1 | Feature flags and analytics | Terra Medium | C2 K1 | Pilot measurement and kill switch |
| M1 | Pre pilot audit | 5.5 High | L1 | Independent adversarial review |
| N1 | Release audit | Sol Ultra | M1 | Whole repo release decision |
| P1A | Video link mode | Terra High | Pilot gate | Conditional mode |
| P1B | Document mode | Terra High | Pilot gate | Conditional mode |
| P1C | Async question mode | Terra High | Pilot gate | Conditional mode |
| P2A | AI source and consent boundary | Sol High | AI gate | Explicit provenance |
| P2B | AI draft recap | Terra High | P2A | Editable draft only |
| P3A | Managed video provider | Sol High | Video gate | Direct upload webhook playback |
| P3B | Meet artifact consent | Sol Extra High | Consent gate | OAuth and artifact discovery |
| P3C | Transcript processing | Sol High | P3B | Async processing retention deletion |

Critical path của P0 là 00 đến 01 đến A1 đến B1 đến B2. Sau B2 có thể làm D1, C1 và E1 tuần tự; không nên chạy song song nếu cùng chạm schema hoặc booking. F, G và H tạo learning loop. I chỉ bắt đầu sau khi workflow artifact đã ổn định. L, M và N là gate bắt buộc trước pilot rộng.

# 8 Prompt triển khai P0

## 8 00 Control plane and baseline

Model 5.6 Sol   Effort High   Phụ thuộc None

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK 00 CONTROL PLANE AND BASELINE
Do not implement Learning Hub features and do not create a schema migration.
Read both supplied DOCX files as reference content, inspect the repository and create:
docs/learning-hub/MASTER_SPEC.md
docs/learning-hub/CONTEXT_PACKET.md
docs/learning-hub/ARCHITECTURE_DECISIONS.md
docs/learning-hub/IMPLEMENTATION_PLAN.md
docs/learning-hub/IMPLEMENTATION_STATE.md
docs/learning-hub/TEST_MATRIX.md
docs/learning-hub/REGRESSION_CHECKLIST.md
docs/learning-hub/CHANGELOG.md
docs/learning-hub/RISK_REGISTER.md
docs/learning-hub/RUNBOOK.md

Preserve LH 001 through LH 100, P0 to P3, out-of-scope items, the persistent LearningSpace decision, mutable
primarySkill, flexible subtopics, no hard pair plus skill unique constraint, three MVP modes and separate
BookingStatus versus FulfillmentStatus.
Record current dirty files without modifying them. Record current routes, schema, scripts, environment variable
names without values, current migrations and high-risk paths. Record current findings that conversations and
messages accept client identity and that auto-complete currently uses CONFIRMED plus endTime after 72 hours;
verify these findings against the latest code.
Run baseline commands without fixing their failures:
npm run db:generate
npx prisma validate
npm run lint
npx tsc --noEmit
npm run build
Run current scripts that are safe with the available test database. Do not fabricate results.
Acceptance:
- Repo memory is concise and internally consistent.
- CONTEXT_PACKET is at most 200 lines.
- Existing behavior and dirty state are unchanged.
- Every P0 LH requirement maps to an implementation task and test placeholder.
Stop after reporting baseline and risks.
```

## 8 01 Minimal test harness

Model 5.6 Terra   Effort Medium   Phụ thuộc 00

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK 01 MINIMAL TEST HARNESS
Build only the smallest repeatable test foundation needed by later Learning Hub tasks.
Reuse the existing package manager and existing scripts. The repository currently has package-lock.json, tsx and
standalone Node test scripts but no detected full test framework; verify this again.
```

```
Prefer node:test plus tsx for TypeScript unit and service tests. Add a heavier framework only if current code cannot be tested meaningfully and explain the cost before doing so.
Add stable package scripts for typecheck, Learning Hub unit tests, integration tests and regression scripts.
Create test helpers for deterministic time, authenticated actors, Prisma mocking or a disposable test database as appropriate. Never point destructive tests at production.
Create a baseline smoke script that checks legacy route/module imports and records environment prerequisites without printing secrets.
Tests for this task:
npm run db:generate
npx prisma validate
npm run typecheck
npm run lint
npm run build
npm run test:learning-hub
Existing standalone regression scripts that can run safely.
Acceptance:
- A failing assertion exits nonzero.
- Tests are deterministic and do not depend on wall clock or external providers.
- No production behavior changes.
- TEST_MATRIX and REGRESSION_CHECKLIST contain exact commands.
```

## 8 A1 Server identity and authorization

Model 5.6 Sol   Effort Extra High   Phụ thuộc 01

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK A1 SERVER IDENTITY AND AUTHORIZATION
Implement only reusable server identity and authorization foundations and close concrete spoofing paths required before private Learning Hub data exists.
Use NextAuth auth() on the server. Add helpers such as requireAuthenticatedUser, requireConversationParticipant and an interface for requireLearningSpaceMember. Do not trust userId, viewerId, senderId or actorId from query, body or client state for authorization.
Audit and fix conversations and messages endpoints first. Preserve legitimate response shapes and chat behavior. Audit booking mutations, notifications and admin boundaries, but only change a concrete vulnerability within this task.
Required tests:
- unauthenticated request is rejected
- user A cannot list user B conversations by query parameter
- nonparticipant cannot open or mark a conversation read
- sender identity comes from session
- legitimate participants list, read and send normally
- admin behavior and booking-to-chat association remain valid
Run auth and chat regression, typecheck, lint and build.
Acceptance:
- No protected chat action authorizes from client identity.
- Existing UI needs only minimal compatibility changes.
- Every changed endpoint documents 401, 403 and 404 behavior.
```

## 8 A2 Realtime and cron security

Model 5.6 Sol   Effort High   Phụ thuộc A1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK A2 REALTIME AND CRON SECURITY
```

```
Implement only realtime privacy and cron authentication.
Audit Pusher channel names, subscription authorization and payloads. Private Learning Hub and conversation
content must use private authorized channels or an equivalent participant guard. Public channels must never
include private notes, filenames, task content or resource metadata.
Enforce CRON_SECRET for sensitive production cron routes. Preserve a clearly isolated local test path only when
NODE_ENV is not production and it cannot be enabled accidentally in production. Do not change settlement timing
or business rules yet.
Required tests:
- unauthorized private channel subscription fails
- nonmember cannot subscribe by guessing an ID
- participant receives legitimate realtime events
- invalid or missing cron secret fails
- valid cron request works
- notification and chat realtime regression passes
- no secret or private payload is logged
Run typecheck, lint, build and relevant route tests.
```

# 8 B1 Learning Hub schema migration

Model 5.6 Sol   Effort Extra High   Phụ thuộc A1 01

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK B1 LEARNING HUB DOMAIN SCHEMA
Implement schema and migration only. Do not add UI, settlement or provider integration.
Add the models and enums approved by MASTER_SPEC: LearningSpace, LearningSpaceMember, LearningTopic,
LearningInvite, LearningResource, LearningTask, Submission, SubmissionReview, LearningNote and LearningActivity.
Add nullable Learning Hub fields to Booking for learningSpaceId, topicId, learningMode, objective snapshot,
definition of done, fulfillment status, deliveredAt and acceptedAt as finalized in the ADR.
Rules:
- LearningSpace persists across bookings.
- Membership does not permanently encode mentor or learner.
- Enforce at most two active members in application service, not an unsafe partial database assumption.
- primarySkill is mutable.
- topics can optionally map to canonical Skill and never auto-publish a Skill.
- no hard unique constraint on pair plus primarySkill.
- old bookings remain valid with null Learning Hub fields.
- define relations, onDelete actions, indexes and audit preservation explicitly.
Create a named migration and rollback note. Do not backfill ambiguous historical rows.
Required verification:
npm run db:generate
npx prisma validate
Apply migration to a clean disposable database.
Apply migration to a representative copy containing legacy bookings.
Run legacy booking queries and build.
Acceptance:
- clean and legacy migration paths pass
- schema supports same pair in multiple spaces
- archived topics and spaces preserve history
- migration does not reinterpret BookingStatus.
```

# 8 B2 LearningSpace services

Model 5.6 Terra   Effort High   Phụ thuộc B1 A1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.
```

```
TASK B2 LEARNINGSPACE DOMAIN SERVICES
Implement domain services and protected handlers for create, read, update and archive of LearningSpace and LearningTopic. No final UI.
Support exactly two active members for MVP, create direct or from a confirmed booking, mutable primarySkill with audit, add rename archive topic, update objective and definition of done with optimistic concurrency, and suggest reuse without forcing it.
Historical booking topic snapshots must not be rewritten when a topic or primarySkill changes. An archived space is readable by its members but rejects new mutations except allowed restore or rebook operations in the spec.
Required tests:
- two active members allowed and third rejected under concurrent requests
- same pair can own multiple spaces
- suggested reuse returns candidates but create still succeeds
- nonmember cannot read or mutate
- optimistic version conflict returns 409 or typed conflict
- topic archive preserves historical references
- primarySkill change creates an audit activity
- old bookings with null space remain readable
Run schema, service, authorization and legacy booking regression.
```

## 8 C1 Bring your pair invite backend

Model 5.6 Terra   Effort High   Phụ thuộc B2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK C1 BRING YOUR PAIR INVITE BACKEND
Implement LH 001 and LH 002 only.
LearningInvite must include inviter, primarySkill, initial objective, tokenHash, expiresAt, usage limit, acceptedAt and revokedAt as applicable. Generate a high-entropy raw token and store only its hash. Return the raw token once.
Acceptance must be transactional and idempotent. Reject self invite, expired or revoked token, unauthorized reuse and any result that would create a third active member. Preserve the intended invite across login or signup using a safe short-lived mechanism.
Required tests:
valid invite, expired invite, revoked invite, replay, concurrent accept, self invite, third member, logged-out continuation, existing active space reuse choice and same pair new-space choice.
Do not change Discover or build social graph features. Do not email until notification task unless an existing safe email helper can be called as best effort without coupling transaction success.
```

## 8 D1 LearningSpace shell

Model 5.6 Terra   Effort Medium   Phụ thuộc B2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK D1 LEARNINGSPACE SHELL
Implement the authorized mobile-first route /learning/[spaceId] using existing GiveGot components and visual patterns.
Show pair, mutable primarySkill, subtopics, shared objective, definition of done, current or next booking, placeholders for resources tasks notes and history, archive state and CTA to book the next session.
Server-authorize before rendering. Do not leak whether a private space exists to a nonmember. Archived spaces stay readable. Long Vietnamese text, empty states, loading, error and keyboard focus states must work.
Do not implement resource, task or note mutations in this task.
Required tests:
member render, nonmember rejection, archived read-only render, empty space, long content, mobile widths, keyboard navigation, dashboard and history regression, no duplicate global navigation.
```

```
Run typecheck, lint and build.
```

## 8 C2 Invite onboarding UI

Model 5.6 Terra   Effort Medium   Phụ thuộc C1 D1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK C2 BRING YOUR PAIR USER EXPERIENCE
Implement the minimum acquisition flow for an existing pair.
Entry point uses clear Vietnamese copy equivalent to Học cùng người bạn đã có. Flow: select primary skill,
optional objective, create link, preview inviter skill and objective, login or signup if needed, accept, open
LearningSpace and present one first-value action.
First-value actions are create first booking, add resource placeholder if F2 is absent, or create task
placeholder if G1 is absent. Hide unavailable actions instead of implementing future tasks.
Handle expired, revoked, already accepted, self-invite and full-space states without exposing private data.
Required tests:
logged-in journey, logged-out continuation, idempotent refresh, invalid token states, back navigation, mobile
layout, copy accessibility and Discover regression.
Instrument only temporary local logs unless L1 analytics contract already exists.
```

## 8 E1 Booking and learning mode integration

Model 5.6 Sol   Effort High   Phụ thuộc B2 D1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK E1 BOOKING AND LEARNING MODE INTEGRATION
Implement only LearningSpace-to-Booking linkage and mode selection for LIVE, EXERCISE_REVIEW and HYBRID.
A booking remains the scheduled or transactional session. LearningSpace remains the persistent pair
relationship. Mode selection must produce mode-specific required-artifact and completion contracts, not only an
enum label.
LIVE requires objective, scheduled booking, meeting link and recap.
EXERCISE_REVIEW requires task, submission and feedback; timing is based on delivery and review window rather
than meeting endTime.
HYBRID requires prework artifact, acknowledgement or questions, scheduled meeting and recap. Prework deadline,
deliveredAt and review window are independent of Meet endTime.
Preserve existing booking creation, AvailableSlot concurrency, Calendar, Meet, cancellation, no-show, dispute
and legacy booking display.
Do not release GP or rewrite cron in this task.
Required tests:
each mode validates its own contract, old booking null mode works, booking links only to member space, topic
snapshot is stable, slot concurrency remains correct, Calendar and Meet creation regress cleanly, cancellation
and dispute states remain reachable.
```

## 8 F1 Private storage infrastructure

Model 5.6 Sol   Effort High   Phụ thuộc B2 A2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and the relevant test and regression sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
database rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract
and stop after this task.

TASK F1 PRIVATE STORAGE INFRASTRUCTURE
```

```
Implement provider abstraction and signed upload/download boundary only. Prefer integration for commodity object storage; do not build storage from scratch.
Use a private bucket. Database stores random storageKey and metadata, never a signed URL. Backend issues short-lived upload or download credentials only after session and active-or-archived membership checks defined by policy.
P0 allowlist should cover PDF, safe images and plain text. Office files wait until malware scanning exists. Enforce MIME, extension sanity, max size, per-space quota and random keys. Do not server-side fetch arbitrary URLs.
Finalize upload metadata only after provider confirmation. Define orphan upload cleanup and soft-delete lifecycle. Keep service keys server-only.
Required tests:
unauthenticated, nonmember, wrong MIME, oversized, quota exceeded, filename traversal, expired URL, deleted resource, orphan finalize, provider failure and successful member upload/download.
Provider calls must be mocked in tests. No test uploads to production.
```

## 8 F2 Learning resources

Model 5.6 Terra   Effort Medium   Phụ thuộc F1 D1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK F2 LEARNING RESOURCE DOMAIN AND USER EXPERIENCE
Implement LH 030, LH 031 and LH 032 on top of F1.
Support HTTPS external links with user-provided title and description and private uploaded files with lifecycle states such as pending, active, deleted and failed. Link a resource to space, optional booking and optional topic.
Do not fetch arbitrary URLs server-side in MVP. Do not parse file content. Do not add raw video upload. Display uploader, type, time, topic and safe status. Members can delete according to policy; deletion must preserve audit and schedule provider cleanup.
Required tests:
link validation including non-HTTPS and dangerous schemes, participant authorization, upload finalize ownership, deleted file denial, quota display, empty/error/mobile states, long Vietnamese filenames, resource activity dedupe and LearningSpace page regression.
Acceptance:
- A pair can add and reopen a link or permitted private file.
- Signed URLs are generated only on demand.
- No signed URL or private bucket path appears in persisted public payloads.
```

## 8 G1 Learning task domain

Model 5.6 Terra   Effort High   Phụ thuộc B2 E1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK G1 LEARNING TASK DOMAIN
Implement LH 040 only. Build task schema behavior, service and protected API or server actions without submission or review UI.
A task belongs to a LearningSpace and may link to a booking and topic. Store creator, assignee, title, description, acceptance criteria, optional dueAt, status and timestamps. Roles come from this task or booking, never permanent LearningSpace member roles.
Validate both users are active members, assignee is eligible, due date is valid and archived spaces reject new tasks. Define state transitions and forbid direct client-controlled terminal states.
Required tests:
create, edit draft, assign, invalid assignee, nonmember, archived space, due date boundary, topic ownership, booking ownership, concurrent edit/version conflict and legacy booking regression.
Do not settle GP or send final notifications yet.
```

## 8 G2 Current submission workflow

Model 5.6 Terra   Effort High   Phụ thuộc G1 F2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK G2 CURRENT SUBMISSION WORKFLOW
Implement LH 041. Support a single current submission per task in MVP with text, safe external HTTPS URL and
allowed LearningResource attachment references.
A first submission records submittedAt. Resubmission after revision increments revisionCount and preserves an
audit event. The API must not let a learner attach resources from another space or impersonate author. Define
exactly when content is editable.
Required tests:
first submit, resubmit, unauthorized author, wrong-space attachment, dangerous URL, empty submission, submit
after task closure, concurrent submit, revisionCount correctness and activity dedupe.
UI must support mobile text and attachment selection, loading, duplicate click prevention and validation
recovery without losing draft.
Do not implement reviewer outcome or GP settlement.
```

## 8 G3 Submission review

Model 5.6 Terra   Effort High   Phụ thuộc G2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK G3 SUBMISSION REVIEW AND REVISION
Implement LH 042. Reviewer can provide private feedback and choose REVIEWED or REVISION_REQUESTED according to
the approved state machine. Feedback is separate from the public Review and trust score.
Only the eligible counterpart can review. A revision request returns task fulfillment to in progress and sets an
optional revision due date. Preserve prior revisions in audit history even if MVP exposes only current content.
Required tests:
valid review, nonreviewer denial, self review denial where roles differ, duplicate review, concurrent review,
revision request, resubmit then review, review after dispute or cancellation, private feedback visibility and
public rating regression.
Create structured activity events but do not send notification if J1 is absent. Do not release GP in this task.
```

## 8 H1 Manual notes and recap

Model 5.6 Terra   Effort Medium   Phụ thuộc D1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK H1 MANUAL NOTES AND RECAP
Implement LH 050. Add private-by-default notes scoped to LearningSpace and optionally Booking. Support note type
GENERAL or RECAP, author, visibility, version and timestamps.
Use optimistic locking for edits. Define delete or archive policy that preserves audit. Do not ingest full chat
and do not send note content to AI.
Required tests:
member create/read/update, nonmember denial, version conflict, archived-space read behavior, deleted note
behavior, booking ownership, long Vietnamese text, XSS-safe rendering, mobile editor and history/dashboard
regression.
Add a concise recap affordance after an eligible session without making it mandatory for legacy bookings.
```

## 8 H2 Structured learning activity

Model 5.6 Terra   Effort Medium   Phụ thuộc F2 G3 H1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK H2 STRUCTURED LEARNING ACTIVITY
Implement LH 060 as an append-only structured feed. Events include space created, topic changed, resource added,
task assigned, submission received, revision requested, feedback ready, note or recap added, booking linked and
fulfillment milestones.
Use stable eventType, entityType, entityId, actorId, timestamp, safe metadata and unique eventKey for dedupe.
Never copy full chat, full note text or private feedback into event metadata.
Required tests:
event creation, retry dedupe, ordering, safe redaction, entity deletion, nonmember access, pagination boundary,
archived history and repeated provider webhook.
UI must have empty, loading and mobile states and avoid an LMS-style complex sidebar.
```

## 8 I1 Fulfillment state machine

Model 5.6 Sol   Effort Extra High   Phụ thuộc E1 G3 H2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK I1 MODE AWARE FULFILLMENT STATE MACHINE
Before coding, audit every current BookingStatus transition, booking mutation, cancellation, no-show, dispute,
review deadline and auto-complete path. Record a transition matrix and ADR.
Implement a separate FulfillmentStatus such as NOT_STARTED, IN_PROGRESS, DELIVERED, REVISION_REQUESTED,
ACCEPTED, DISPUTED and CANCELLED as finalized by the spec. Do not reinterpret BookingStatus.
Eligibility:
LIVE follows live attendance and review policy.
EXERCISE_REVIEW reaches DELIVERED only after required feedback exists.
HYBRID reaches DELIVERED only after required prework, scheduled meeting component and recap are complete. Meet
endTime alone is insufficient.
Use validated service transitions, conditional updates and audit events. Client cannot set arbitrary state.
Required tests cover every allowed and forbidden transition, retry, stale version, concurrent deliver and
accept, cancellation, dispute freeze, revision loop, old null fulfillment booking and Hybrid meeting ending
before prework completion.
Do not release GP or rewrite cron yet.
```

## 8 I2 GivePoint settlement

Model 5.6 Sol   Effort Extra High   Phụ thuộc I1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md,
ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git
status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old
rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop
after this task.

TASK I2 ATOMIC GIVEPOINT SETTLEMENT
Implement settlement triggered only by an eligible ACCEPTED fulfillment or the approved mode-aware timeout. Use
one database transaction and conditional claim so double click, retry, concurrent request and cron cannot credit
twice.
Preserve existing TransactionLog compatibility. Add an idempotency key or equivalent unique invariant for
completion settlement. Never infer price from duration unless an approved business rule exists. Pilot may remain
one GP per accepted loop, but do not market it as hourly pay.
Dispute, cancellation, MISSED and unresolved revision must block settlement. Notification failure is post-commit
best effort and cannot roll back a successful ledger transaction.
```

```
Required tests:
manual accept, concurrent accept, duplicate retry, cron race, dispute race, cancellation, revision requested, wrong actor, legacy live booking, Exercise Review deliveredAt timeout, Hybrid incomplete artifacts, transaction rollback, notification failure and wallet plus ledger reconciliation.
Add a reconciliation query or script that finds completed bookings without exactly one matching successful completion credit.
Do not change cron selection in this task.
```

## 8 I3 Mode aware cron

Model 5.6 Sol   Effort High   Phụ thuộc I2 A2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK I3 MODE AWARE AUTO SETTLEMENT CRON
Replace broad CONFIRMED plus endTime selection with explicit mode-aware eligibility while preserving legacy Live policy where required.
LIVE may use the approved endTime review window.
EXERCISE_REVIEW review window starts at deliveredAt.
HYBRID timeout starts only when the complete required artifact checklist reaches DELIVERED, not when Meet ends.
Exclude DISPUTED, CANCELLED, MISSED, REVISION_REQUESTED, already claimed and incomplete rows. Enforce CRON_SECRET and use a deterministic injected clock for tests. Process in bounded batches and report counts without private content.
Required tests:
boundary before and at deadline, timezone independence, each mode, incomplete Hybrid, revision reset, dispute, duplicate cron, two cron workers, partial transaction failure, notification failure and legacy booking policy.
Run the existing review-deadlines and PA-02 scripts to prove no-show and review policy still work.
```

## 8 J1 Learning Hub notifications

Model 5.6 Terra   Effort Medium   Phụ thuộc G3 I2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK J1 LEARNING HUB NOTIFICATIONS
Implement LH 090 using the existing Notification model and email helper where appropriate. Events include invite, resource added, task assigned, due soon, submission received, feedback ready, delivery accepted and auto settlement.
Use an event-derived dedupe key or outbox-compatible pattern so retries do not spam. Domain transaction success must not depend on email or realtime provider success. Deep links must authorize the destination.
Do not include note content, feedback text, filenames or private resource URLs in public realtime payloads or email subject lines.
Required tests:
one notification per event, retry dedupe, provider failure, deep-link authorization, wrong recipient, due boundary, read state regression, existing booking notification regression and no private payload leak.
```

## 8 K1 Learning history and rebook

Model 5.6 Terra   Effort Medium   Phụ thuộc H2 I3

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK K1 LEARNING HISTORY AND REBOOK
```

```
Implement LH 080. Show persistent LearningSpace history across bookings, topics, safe artifact summaries, fulfillment milestones and archived content. Provide rebook from the relationship without duplicating the space by default; still let the user create another space when intentional.
Old bookings without a LearningSpace continue to appear in existing history. Archived spaces remain readable but mutation rules stay enforced.
Required tests:
multi-booking timeline, topic changes, archived topic, archived space, old null-space booking, pagination, rebook preserving space, explicit new-space choice, nonmember denial, mobile view and existing history regression.
```

## 8 K2 Contribution evidence

Model 5.6 Terra   Effort Medium   Phụ thuộc K1 I2

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK K2 CONTRIBUTION EVIDENCE
Implement P1-safe aggregates from authoritative records: settled completed sessions, confirmed contributed duration, accepted assignment reviews and existing public rating policy. Decide visibility per field; private resource, note and feedback content is never public.
Do not recompute GP from hours. GP earned comes from TransactionLog if product policy exposes it. Avoid double counting bookings, retried events or revised submissions.
Required tests:
aggregation correctness, no double count, null legacy data, hidden review, archived space, timezone duration, privacy by viewer, pagination and reconciliation with wallet or ledger.
Keep the UI modest and evidence-oriented; do not build certificates, grades or badges.
```

## 8 L1 Feature flags and product analytics

Model 5.6 Terra   Effort Medium   Phụ thuộc C2 K1

```
Do not rely on previous chat context. First read docs/learning-hub/MASTER_SPEC.md, CONTEXT_PACKET.md, ARCHITECTURE_DECISIONS.md, IMPLEMENTATION_STATE.md and relevant test, regression and risk sections. Inspect git status and preserve all pre-existing user changes. Work only on this task. Preserve legacy GiveGot flows and old rows. Use server-session identity. Do not refactor unrelated code. Apply the Task Completion Contract and stop after this task.

TASK L1 FEATURE FLAGS AND PRODUCT ANALYTICS
Add a kill switch and pilot allowlist for Learning Hub without breaking normal GiveGot navigation. Define server-validated flag behavior and safe fallback.
Implement analytics events from the master spec: learning_invite_created, learning_invite_accepted, learning_first_value_completed, learning_loop_completed, learning_space_returned and learning_rebooked. Use stable schemas, anonymous or minimum identifiers, dedupe and consent policy. Do not send note, task, feedback or resource content.
Define activation as both members present plus first useful artifact or booking within the approved window. Define week 2 and week 4 retention precisely.
Required tests:
flag off, pilot user on, nonpilot off, event schema, dedupe, no private content, retry, activation boundary and existing analytics or navigation regression.
Add a pilot dashboard query or documented SQL, not a large admin BI product.
```

# 9 Prompt audit và pilot

## 9 M1 Pre pilot adversarial audit

Model 5.5   Effort High   Phụ thuộc L1

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass. Apply the Task Completion Contract and stop.
```

```
TASK M1 PRE PILOT ADVERSARIAL AUDIT
Perform an independent audit. Do not implement new roadmap features and do not silently fix findings.
Check every P0 LH requirement against code and tests. Trace these complete journeys:
1. existing pair invite to accepted LearningSpace to first value
2. Live booking to recap to accepted settlement
3. Exercise Review task to submission to feedback to revision or acceptance
4. Hybrid prework to scheduled meeting to recap to acceptance
5. cancellation no-show dispute and admin support
Audit authorization, object-level access, signed URLs, Pusher payloads, cron secret, migration compatibility,
idempotency, concurrency, notification retry, mobile and accessibility.
Run the full command suite and reconciliation scripts with a disposable database. Produce docs/learning-
hub/PRE_PILOT_AUDIT.md with severity, evidence, reproduction, affected requirement, suggested smallest fix and
release blocker flag.
Acceptance:
- no claim without file, test or runtime evidence
- every P0 requirement is PASS, FAIL or NOT TESTED
- no code changes except audit artifacts
- recommendation is GO, CONDITIONAL GO or NO GO.
```

## 9 N1 Final release audit

Model 5.6 Sol   Effort Ultra   Phụ thuộc M1 blockers fixed

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK N1 FINAL RELEASE AUDIT
Read PRE_PILOT_AUDIT.md and verify each release blocker was fixed by a dedicated task. Audit the whole
repository at the latest checkpoint.
Evaluate product requirement compliance, architecture, backward compatibility, authentication, authorization,
storage, LearningSpace consistency, booking, learning modes, fulfillment, GP settlement, transaction
idempotency, concurrency, cron, notifications, migration, legacy data, chat, Calendar, Meet, reviews, trust,
wallet, admin, responsive behavior, accessibility, observability and test coverage.
Run clean-install verification using package-lock.json, database generation and validation, full test suite,
lint, typecheck and production build. Run a clean database migration and a representative legacy-data migration.
Run E2E journeys where environment permits; mark unavailable checks honestly.
Do not add features. Only make a trivial documentation correction if it cannot affect production; otherwise
report a blocker and stop.
Produce RELEASE_AUDIT.md, RELEASE_CHECKLIST.md and a rollback checklist. Release recommendation requires zero
unresolved critical or high security, settlement or migration findings.
```

## 9 3 Prompt sửa blocker

Dùng prompt này cho từng finding riêng. Không gom nhiều finding không cùng root cause.

```
REPAIR ONE VERIFIED LEARNING HUB FINDING
Finding ID: [PASTE ID]
Evidence and reproduction: [PASTE FROM AUDIT]
Required expected behavior: [PASTE]
Allowed files or modules: [LIST]
Do not rely on previous chat. Read MASTER_SPEC, CONTEXT_PACKET,
IMPLEMENTATION_STATE and the audit finding. Inspect the latest code.
Reproduce the failure before changing code. Implement the smallest root-cause fix.
Add a regression test that fails before and passes after the fix.
Run the task-specific suite and all regressions named by the finding.
Do not refactor adjacent code or fix other findings.
Update state, test matrix and changelog. End with the Completion Contract status.
```

## 9 4 Pilot gate

| Gate | Khuyến nghị tối thiểu trước khi mở rộng |
|---|---|
| Security | Không còn critical hoặc high object-level authorization, secret hoặc private-file finding. |
| Settlement | Reconciliation bằng không; retry và concurrent cron không tạo duplicate credit. |
| Migration | Clean DB và representative legacy data đều migrate và rollback plan được diễn tập. |
| Product | 5 đến 10 pair hoàn tất onboarding; ít nhất 5 learning loops hoàn chỉnh. |
| Differentiation | Có pair thực sự dùng Exercise Review hoặc Hybrid, không chỉ Live. |
| Reliability | Không có stuck fulfillment không có runbook; provider failure không làm mất domain state. |
| Measurement | Activation, loop completion, rebook, week 2 và week 4 definitions đã cố định. |

Các con số pilot là ngưỡng vận hành khuyến nghị, không phải dữ kiện đã được chứng minh. Owner có thể điều chỉnh và phải ghi thay đổi vào ADR hoặc IMPLEMENTATION_STATE trước khi mở gate.

# 10 Prompt mở rộng có điều kiện

Các prompt trong phần này nằm ngoài P0. Mỗi prompt chỉ được chạy sau khi gate tương ứng được owner xác nhận bằng dữ liệu pilot.

## 10 P1A Video link learning mode

Model 5.6 Terra   Effort High

Gate Có ít nhất ba pair yêu cầu và external video link đã được dùng lặp lại

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P1A VIDEO LINK LEARNING MODE
Add a VIDEO mode using external HTTPS video links only. Do not add raw upload.
Define required objective, delivered video resource, question or acknowledgement and acceptance. Use deliveredAt
and a review window; never settle from a synthetic meeting endTime.
Reuse private resources, activity, notifications and fulfillment services. Restrict supported hosts by safe
policy or store user metadata without server-side fetching.
Test allowed and denied URLs, member access, delivery, acceptance, revision or questions, timeout, dispute,
duplicate settlement, old modes and mobile UI.
```

## 10 P1B Document learning mode

Model 5.6 Terra   Effort High

Gate Users repeatedly exchange reading material and request a document workflow

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P1B DOCUMENT LEARNING MODE
Add DOCUMENT mode on top of private LearningResource, task and notes. Required artifacts are a permitted
document or external document link, reading objective, expected learner response and optional provider feedback.
Do not build collaborative document editing, PDF annotation or malware scanning in this task. Office upload
remains disabled unless scanning is already operational.
Completion and settlement follow deliveredAt plus accepted response, not scheduled time.
Test document access, response requirement, revision, timeout, dispute, unsupported Office upload, legacy modes
and private-content boundaries.
```

## 10 P1C Async question and answer mode

Model 5.6 Terra   Effort High

Gate Pilot shows repeated asynchronous questions that do not fit Exercise Review

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P1C ASYNC QUESTION AND ANSWER MODE
Add ASYNC_QA with question, answer, clarification thread and learner-resolved state. Use structured domain
records; do not reuse unrestricted chat as the source of settlement truth.
Define who may answer, when clarification reopens the item, delivery timestamp, resolve window, dispute and
settlement eligibility.
Test unauthorized reader, answer delivery, reopen, resolve, timeout, duplicate resolution, dispute race,
notification dedupe and chat regression.
```

## 10 P2A AI source consent and provenance

Model 5.6 Sol   Effort High

Gate Notes and recap usage is sustained and users request AI assistance

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P2A AI SOURCE CONSENT AND PROVENANCE
Build only the AI safety and provenance boundary. Do not generate summaries yet.
A member explicitly selects source notes, task records, feedback or transcript segments. The server verifies
access to every source at execution time, records consent, source IDs, model metadata, prompt version, creator
and retention status. Full chat, arbitrary URL content and private files are excluded unless a later approved
parser exists.
Define deletion propagation, opt-out, retry and cost limits. Never expose provider keys.
Test cross-space source injection, deleted source, revoked consent, nonmember, prompt injection content
isolation, retry, audit metadata and zero-source request.
```

## 10 P2B AI draft recap

Model 5.6 Terra   Effort High

Gate P2A passes security review and AI gate is approved

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P2B AI DRAFT RECAP
Use the existing Gemini integration behind a feature flag to generate an editable DRAFT recap from explicitly
selected sources only.
Output must show provenance, generation time and draft status. A human must review and publish; AI cannot mark
fulfillment complete, change ratings, settle GP or create authoritative facts. Preserve the original manual
notes.
Use bounded input, timeout, retry, cost telemetry and safe error states.
Test grounded-source selection, inaccessible source, hallucination-resistant empty evidence behavior, prompt
injection content, edit before publish, opt-out, deletion, provider failure and no side effects on fulfillment
or settlement.
```

## 10 P3A Managed video provider

Model 5.6 Sol   Effort High

Gate Raw video demand justifies provider cost and operations

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P3A MANAGED VIDEO PROVIDER
Integrate Mux or Cloudflare Stream after an ADR compares cost, upload, webhook, signed playback, retention and
regional requirements. Do not build a custom encoder or proxy large video through the app server.
Use direct creator upload, verified idempotent webhook, provider asset lifecycle and signed playback after
membership authorization. Record processing, ready, failed and deleted states.
Test forged webhook, replay, wrong-space playback, expired token, abandoned upload, processing failure,
deletion, quota and provider outage. Add orphan and stuck-processing runbook procedures.
```

## 10 P3B Google Meet artifact consent

Model 5.6 Sol   Effort Extra High

Gate Workspace account support scopes consent and privacy review are approved

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P3B GOOGLE MEET ARTIFACT CONSENT
Implement OAuth scope and consent handling for discovering Meet recordings or transcripts. Do not request
broader scopes than required and do not auto-import artifacts without explicit member action and policy
approval.
Map a Meet conference to an eligible booking, verify participant and space membership, record consent and
provider artifact metadata, and handle unavailable artifacts.
Test revoked OAuth, missing scope, wrong conference, nonmember, artifact not ready, duplicate import, consent
withdrawal and existing Calendar or Meet creation regression.
```

## 10 P3C Transcript processing and retention

Model 5.6 Sol   Effort High

Gate P3B approved and transcript retention policy is signed off

```
Do not rely on previous chat context. Read the repository Learning Hub source of truth and latest implementation
state. Inspect git status and preserve user changes. Do only this task. Do not weaken tests to obtain a pass.
Apply the Task Completion Contract and stop.

TASK P3C TRANSCRIPT PROCESSING AND RETENTION
Process approved transcript artifacts asynchronously. Store only required metadata and content under the
approved retention and deletion policy. Access remains private and all derived AI use requires the P2A source-
consent boundary.
Implement job idempotency, status, retries, dead-letter or support path, deletion propagation and audit. Do not
auto-complete fulfillment from transcript availability.
Test duplicate job, partial artifact, revoked consent, deletion, wrong-space access, provider outage, retry
exhaustion, privacy export and no settlement side effect.
```

# 11 Regression matrix

| Flow cũ | Invariant phải giữ | Task nhạy cảm | Bằng chứng tối thiểu |
|---|---|---|---|
| Authentication | Google và credentials login, session user id, suspension policy hoạt động | A1 A2 | Auth tests và route smoke |
| Discover và matching | Search semantic và profile discovery không đổi | C2 D1 | Build và targeted UI regression |
| Mentor profile | Public data và booking CTA không đổi | D1 K2 | Profile render test |
| AvailableSlot | Một slot không bị double book | E1 I1 | Concurrent booking test |
| Booking lifecycle | PENDING CONFIRMED COMPLETED CANCELLED MISSED DISPUTED còn đúng nghĩa | E1 I1 I3 | Transition matrix và legacy fixtures |
| Google Calendar | Event tạo sửa hủy theo flow hiện tại | E1 I3 | Mock integration regression |

| Flow cũ | Invariant phải giữ | Task nhạy cảm | Bằng chứng tối thiểu |
|---|---|---|---|
| Google Meet | Meeting link creation không bị coi là mọi fulfillment | E1 I1 | Live and Hybrid tests |
| Cancellation | Refund và slot release đúng policy | I1 I2 I3 | Cancellation scripts |
| No show and dispute | Funds freeze và admin resolution giữ nguyên | I1 I2 I3 | PA 02 and dispute cases |
| Wallet and ledger | Số dư khớp TransactionLog, không double credit | I2 I3 K2 | Reconciliation script |
| Public reviews | Submission feedback không làm thay đổi public Review | G3 K2 | Review visibility regression |
| Trust score | Không đổi từ private task feedback | G3 K2 | Trust history assertion |
| Chat | Participant đọc gửi realtime bình thường | A1 A2 H2 | Chat route and Pusher tests |
| Notifications | Legacy notification read state và deep link hoạt động | A2 J1 | Notification regression |
| Admin | Role guard và support flow không mở rộng dữ liệu mặc định | A1 I2 | Admin authorization tests |
| AI quiz and roadmap | Không bị phụ thuộc Learning Hub | D1 L1 | Route build smoke |
| Dashboard and history | Legacy rows null LearningSpace vẫn hiển thị | B1 K1 | Legacy data fixture |
| Cron jobs | Secret, batch, retry và idempotency đúng | A2 I3 | Deterministic clock tests |

# 12 Lệnh kiểm thử và xử lý failure

## 12 1 Test ladder

| Mức | Lệnh hoặc loại kiểm thử | Khi chạy |
|---|---|---|
| T0 | npm run db:generate và npx prisma validate | Mọi schema hoặc Prisma client change |
| T1 | npm run typecheck và targeted node:test qua tsx | Mọi code task |
| T2 | Service route integration với mock hoặc disposable database | Domain auth storage settlement |
| T3 | Standalone regression scripts liên quan | Mọi task chạm flow cũ |
| T4 | npm run lint và npm run build | Cuối task UI API hoặc cross-module |
| T5 | Full Learning Hub suite migration matrix E2E reconciliation | Cuối block, pre-pilot và release |

Task 01 phải ghi exact script names vào package.json và TEST_MATRIX.md. Các prompt sau dùng tên đã được tạo thay vì tự phát minh command mới.

```
CANONICAL COMMAND SET AFTER TASK 01
npm ci
npm run db:generate
npx prisma validate
npm run typecheck
npm run test:learning-hub
npm run test:learning-hub:integration
npm run test:regression
npm run lint
npm run build

Run npm ci only for clean-install gates, not repeatedly in every task.
Database-writing tests must use an explicit disposable DATABASE_URL.
Never print secrets or run destructive fixtures against production.
```

## 12 2 Mức test tối thiểu theo task

| Task | Mức bắt buộc | Audit trọng tâm |
|---|---|---|
| 00 | Baseline T0 T4 | Ghi failure nhưng không sửa |
| 01 | T1 và self-test runner | Nonzero exit and isolation |
| A1 A2 | T1 T2 T3 T4 | Spoofing object access secret payload |
| B1 | T0 T2 T3 T4 | Clean and legacy migration rollback |
| B2 C1 | T1 T2 T3 | Concurrency membership token |
| C2 D1 | T1 T3 T4 plus UI | Mobile accessibility legacy routes |
| E1 | T0 T1 T2 T3 T4 | Booking slot Calendar Meet |
| F1 F2 | T1 T2 T3 T4 | Signed URL MIME quota orphan |
| G1 G2 G3 | T1 T2 T3 T4 | State role revision privacy |
| H1 H2 | T1 T2 T3 T4 | Optimistic lock dedupe redaction |
| I1 I2 I3 | T0 to T5 | Transition race ledger reconciliation |
| J1 | T1 T2 T3 T4 | Dedupe failure and private payload |
| K1 K2 | T1 T2 T3 T4 | Legacy history aggregate privacy |
| L1 | T1 T2 T3 T4 | Flag server enforcement analytics privacy |
| M1 N1 | T0 to T5 | Evidence-based release decision |

## 12 3 Failure protocol

1. Reproduce và ghi command, exit code, failing assertion cùng fixture.
2. Xác định failure có tồn tại ở baseline Prompt 00 hay do task mới.
3. Nếu do task mới, task không được PASS. Sửa trong cùng scope hoặc báo BLOCKED.
4. Nếu pre-existing nhưng cản test bắt buộc, mở prompt riêng để sửa baseline; không giấu bằng skip hoặc đổi assertion.
5. Không xóa, weaken, snapshot-update hoặc mock bỏ behavior chỉ để đạt màu xanh.
6. Sau fix chạy targeted test trước; chỉ chạy lại full suite khi targeted pass hoặc task là block gate.
7. Ghi limitation cùng owner, impact và follow-up ID. PASS WITH KNOWN LIMITATION không được dùng cho security, settlement hoặc migration blocker.

## 12 4 Quy tắc chia nhỏ task

- Tách task nếu đồng thời chạm hai vùng high risk như auth và settlement, migration và UI, hoặc storage và AI.
- Tách task nếu cần hơn một migration độc lập, hơn hai domain aggregates hoặc không thể mô tả rollback trong một đoạn ngắn.
- Tách audit khỏi fix. Audit tạo finding; mỗi finding có prompt sửa riêng và regression test riêng.
- Không để Codex tự triển khai task tiếp theo vì còn quota hoặc context. Stop condition là bắt buộc.
- Một task nên có một outcome có thể demo, một owner của state transition và một tập regression hữu hạn.

# 13 Checklist vận hành thực tế

## 13 1 Trước mỗi task

- Task phụ thuộc đã PASS và checkpoint có thể đọc từ checkout hiện tại.
- git status được xem trước; user changes được liệt kê và bảo toàn.
- Model và effort khớp registry; không dùng Ultra nếu không phải N1.
- Chỉ copy một task prompt; không dán lại toàn bộ DOCX vào conversation.
- Codex đọc repo memory và relevant source files thay vì quét lại toàn repo không mục tiêu.
- Environment test không trỏ production; provider test dùng mock.

## 13 2 Sau mỗi task

- Đối chiếu acceptance từng dòng và xem diff trước khi tin summary.
- Xác nhận test command thật sự chạy, không chỉ được đề xuất.
- Kiểm tra migration, route contract, error state và backward compatibility nếu có.
- Xác nhận IMPLEMENTATION_STATE và CONTEXT_PACKET đã cập nhật ngắn gọn.
- Stage explicit paths. Tạo commit hoặc giữ cùng checkout theo workflow đã chọn.
- Chỉ chuyển task tiếp theo khi status PASS; limitation phải có follow-up owner.

## 13 3 Cách mở task Codex mới

1. Chọn đúng GiveGot project và bắt đầu từ branch hoặc working tree chứa checkpoint gần nhất.
2. Chọn model và effort theo registry.
3. Dán nguyên prompt của task tương ứng. Không thêm yêu cầu feature khác vào cuối prompt.
4. Nếu Codex phát hiện conflict lớn, để nó dừng và trả decision request. Trả lời bằng quyết định cụ thể rồi tiếp tục cùng task.
5. Khi task hoàn tất, review diff và test evidence. Sau đó mới tạo task Codex tiếp theo.

## 13 4 Prompt kiểm tra trạng thái nhanh

```
READ ONLY LEARNING HUB STATUS CHECK
Do not change files.
Read docs/learning-hub/IMPLEMENTATION_STATE.md, CONTEXT_PACKET.md,
TEST_MATRIX.md and git status.
Report:
- latest PASS checkpoint
- current dirty files and whether they belong to the last task
- next dependency-ready task
- unresolved blocker or owner decision
- exact model and effort for the next task
- exact tests that next task must preserve
Do not propose or implement additional features.
```

## 13 5 Prompt đồng bộ context khi drift

```
CONTEXT RECONCILIATION ONLY
Do not change production code.
Compare MASTER_SPEC, ARCHITECTURE_DECISIONS, IMPLEMENTATION_STATE,
CONTEXT_PACKET, TEST_MATRIX and the latest source code.
Identify stale or contradictory statements with file evidence.
Update repository documentation only after proving the current code state.
```

```
Do not reinterpret business policy, rewrite history or mark untested work PASS.
Keep CONTEXT_PACKET at most 200 lines and record the reconciliation in CHANGELOG.
```

### 13 6 Những yêu cầu nên thêm khi cần

| Tình huống | Câu bổ sung vào prompt |
|---|---|
| Có dirty worktree | Treat all pre-existing changes as user-owned. Do not restore, reset, reformat or stage them. |
| Chạm payment hoặc GP | Show the ledger invariant and prove idempotency under concurrent calls before coding. |
| Chạm migration | Test clean and representative legacy databases and document rollback before applying to a shared environment. |
| Chạm provider | Mock provider in tests, verify webhook signatures and keep domain transaction independent from provider availability. |
| Chạm privacy | Enumerate actor, object and action authorization for every endpoint and signed URL. |
| Chạm UI | Test empty loading error mobile keyboard focus and long Vietnamese content states. |
| Quota thấp | Read only files relevant to this task, run targeted tests first and stop when the Completion Contract is satisfied. |
| Context nghi ngờ stale | Run Context Reconciliation Only before any feature change. |

## Phụ lục quyết định sản phẩm phải giữ

- LearningSpace là relationship bền vững của một pair, có primarySkill để định vị và subtopics linh hoạt.
- Không có hard unique constraint pair plus primarySkill; hệ thống gợi ý reuse nhưng user có thể tạo space riêng.
- Role mentor hoặc learner thuộc booking hay task cụ thể, không khóa vĩnh viễn ở membership.
- MVP chỉ gồm Live, Exercise Review và Hybrid. Mode phải thay đổi checklist, artifacts và completion rules.
- Hybrid có meeting theo lịch nhưng prework, deliveredAt, review window và settlement không phụ thuộc riêng endTime.
- BookingStatus và FulfillmentStatus giải quyết hai vấn đề khác nhau.
- Private by default, server session owns identity, signed URLs có TTL ngắn và được cấp sau authorization.
- Build workflow, permissions, fulfillment, reputation và history trong GiveGot; tích hợp storage, meeting, email và video provider.
- AI chỉ tạo draft từ nguồn user chọn, có provenance, consent và human review; không có quyền settlement.
- Mọi rollout dùng feature flag, pilot nhỏ, analytics tối thiểu và rollback runbook.